Mind-Reading Readiness: Trying to Predict BCI Literacy with Minimal EEG Features

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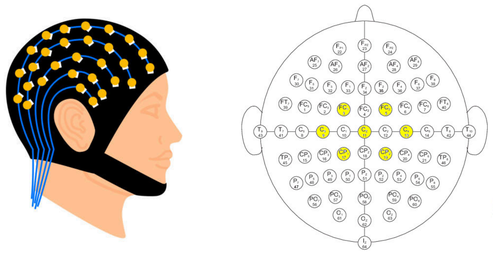
Background + Goal

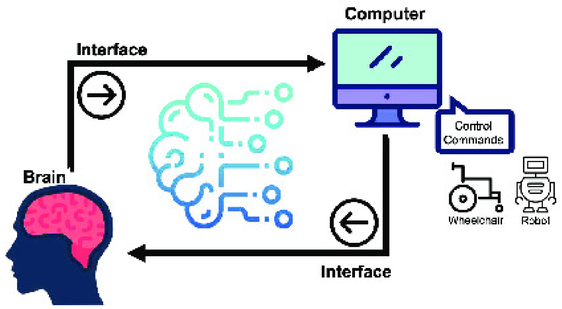
- **Motor Imagery Brain-Computer Interfaces (MI BCIs)** are machines that use brain activity data (like from EEG tests) to perform actions
- **BCI literacy (ability to use a BCI)** varies among people, testing for it is often expensive and time-consuming
- **EEG readings** have predictive value for BCI literacy, and can be relatively easy, and cheap to obtain (Blankertz et al. (2010))

Our goal: Use exclusively EEG features and limited recordings to identify overall BCI illiteracy

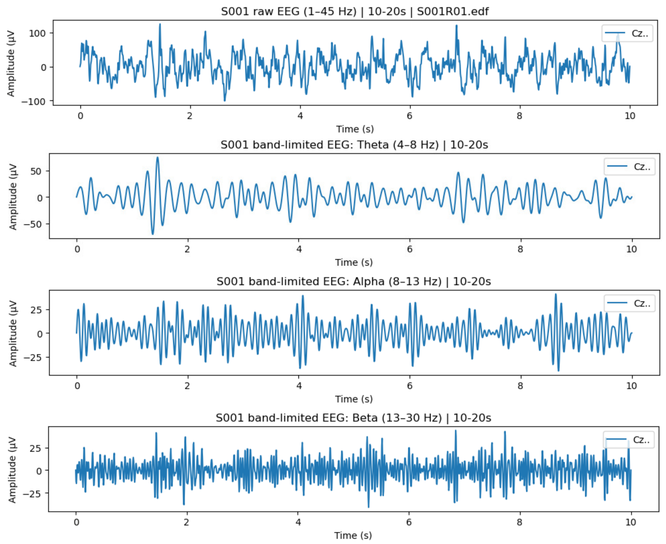
EEG headset example and 64-channel template (motor cortex electrodes highlighted)



How a BCI works



Examples of raw and frequency band-limited EEG recordings, taken from Subject 001 from the Physionet EEG Dataset, used Cz electrode



Methods Overview

Data (PhysioNet MI EEG)

109 subjects • 64-ch EEG • 160 Hz (BCI2000)
Rest (R01/R02) + MI L/R imagery (runs 4/8/12)
ROI: C3/Cz/C4 (sensorimotor focus)

Preprocess + Epoch

MNE: filter, re-reference, artifact handling
MI epochs: baseline −1–0 s, task 1–4 s
Consistent windows for all features

Ground Truth Literacy

Within-subject MI decoder: CSP → LDA
5-fold CV accuracy per subject = target
Used only for labels (not inputs)

Feature Merge (Low-Sample)

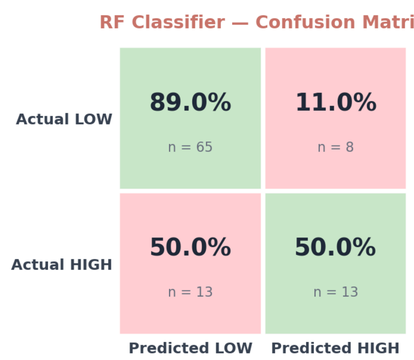
Early MI (first N=15): power, ERD/ERS, CSP-sep, SNR/var
Rest predictors + rhythm metrics (SMR/RPL, TAR, PSE)
Merge → 38 features; median impute if missing

Models + Evaluation

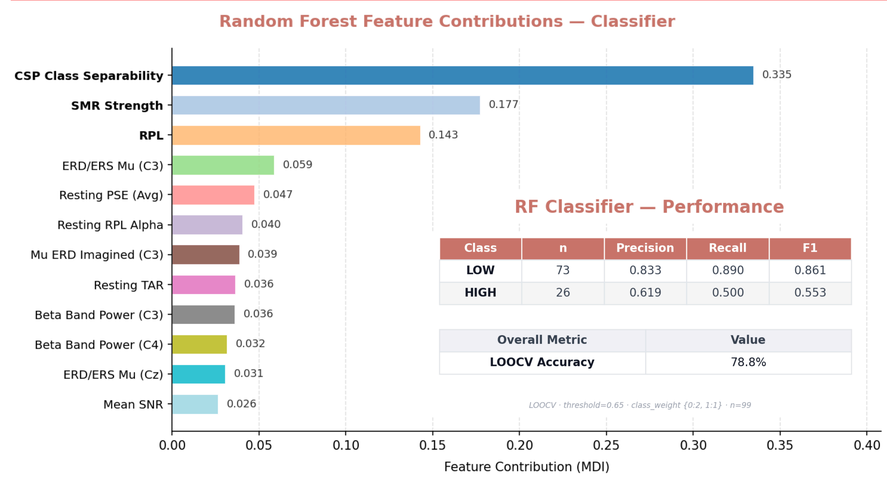
Regression: compare RF/GB/SVR/Ridge → best RF
Binary RF classifier: high vs low (threshold = 0.65)
Report CV/LOOCV + final feature importance

Binary RF Classifier Confusion Matrix (total n=99)

Can accurately guess for BCI-Illiterate subjects (guessed correctly for Actual LOW 89%) but struggles with predicting for high literacy subjects (only 50% accuracy for guessing Actual HIGH)



RF Binary Classifier Model Performance Table and Feature Weights



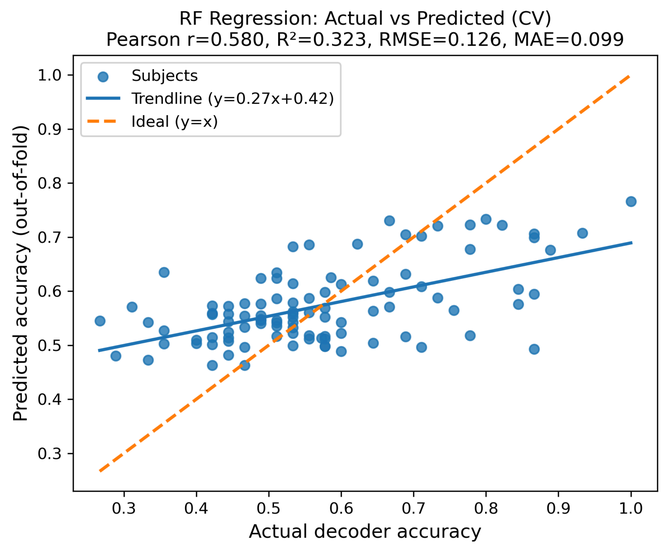
RF Classifier — Performance

Class	n	Precision	Recall	F1
LOW	73	0.833	0.890	0.861
HIGH	26	0.619	0.500	0.553

Overall Metric	Value
LOOCV Accuracy	78.8%

LOOCV · threshold=0.65 · class_weight {0:2, 1:1} · n=99

RF Regression Model Actual vs Predicted Plot



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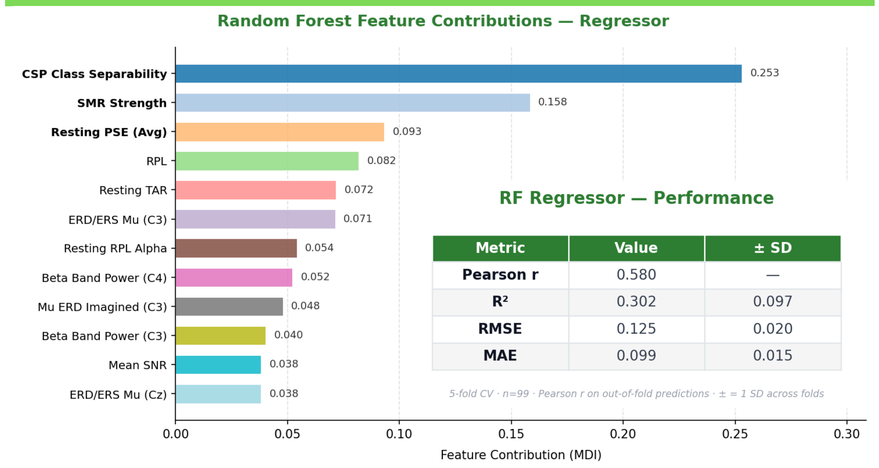

RF Regressor — Performance

Metric	Value	± SD
Pearson r	0.580	—
R^2	0.302	0.097
RMSE	0.125	0.020
MAE	0.099	0.015

5-fold CV · n=99 · Pearson r on out-of-fold predictions · ± = 1 SD across folds

Results

- **Final Random-Forest Binary Classification Model:** 78.8% accurate, 89% of low performers correctly classified
- **Final Random-Forest Regression Model:** Pearson's r: 0.58, r-squared = 0.30, Spearman r: 0.47, moderate positive association

Conclusions

- EEG resting-state and MI trial features are **statistically significant** in predicting MI BCI compatibility
- **Neither alone is sufficient** for relaible classification
- Our features were enough to **reliably predict** for **BCI iliteracy** but were **not sufficient for specificity** (regression), likely need other features
- Future work could focus on using non-EEG but still easy-to-collect features and/or validating on other populations

Resources, Data Collections

- **Physionet EEG Motor Movement /Imagery Dataset** (Goldberger et al. (2000))
- **MetaBCI** (Mei et al. (2024))
- Larson, E., et al. MNE-Python. v1.11.0, Zenodo, 21 Nov. 2025,